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Artificial intelligence-based chatbots – a motivation underlying sustainable development in banking: standpoint of customer experience and behavioral outcomes

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ABSTRACT

In the digital transformation scenario, banks must strengthen business sustainability through digital technologies, such as artificial intelligence-based chatbots. Crucial evidence illustrates that using chatbots allows banks to enhance business performance and bank–customer relationship. This study aims to unveil how chatbots bring customer experiences and whether customers adopt behavioral outcomes. Structural equation modeling is calculated to examine the hypotheses of a postulated research model with a valid sample of 336 respondents. The results reveal that extrinsic value is investigated because of the central indispensability of chatbot attributes, including understandability, automation, and competence. Meanwhile, understandability, personalization, interaction, and intimacy are of paramount importance in increasing the intrinsic value toward chatbot usage. Intrusiveness tends to inhibit customers' intrinsic value. Furthermore, extrinsic and intrinsic values significantly predict customer satisfaction and continuance intention toward banking chatbots. Additionally, satisfaction is a motivation for customer intention to sustain the use of banking chatbots. In light of the obtained findings, it is imperative for banks to deem chatbot characteristics as the determinative motivations for customer experience because chatbots provide customers with valuable information and have the capability to address financial issues.

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1. Introduction

Digital transformation has pushed the diffusion of digital technologies forward, leading to varied business models for enhancing customer experience and dealing with information and services (T. H. Nguyen et al., 2022). The chatbot is one such salient artificial intelligence-based digital technology. It serves as a virtual agent that can replace humans and conduct natural, automatic communication with users (Mostafa & Kasamani, 2022). Chatbots have become an increasingly powerful means for banks to promote effective customer interaction (D. M. Nguyen et al., 2021), consequently leading to the improvement of service quality, cementing of bank–customer relationships (Le, 2023), and increase of online customer experience (Chen et al., 2021).

Vietnam banks have willingly accepted a strong wave of digital transformation in the last 3 years (Ha & Nguyen, 2022), moving toward the integration of digital technology solutions, including chatbots, to contribute to modern business models and sustainable strategies for banking development (D. M. Nguyen et al., 2021). A report by Ha and Nguyen (2022) showed that 31.9% of banks have implemented digital transformation in banking services, and 70% have expressed their readiness to utilize digital technologies. By 2022, 14/43 banks (corresponding to 32.5%) have already implemented the usage of chatbots (Huong & Binh, 2022). Chatbots are beneficial to banks when it comes to running marketing campaigns and customer relationship management and offering personalized customer services (D. M. Nguyen et al., 2021). Chatbots can converse with customers and give them

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instructions because they are capable of performing banking operations comprising iterative processes with stringent standards, such as greetings, making consultation appointments, collecting reviews and inquiries, delivering information, and handling financial transactions. This indispensability tempts banks to cogitate how chatbots should be designed to foster customer experience and engagement.

Despite the acknowledgement of the potential of banking chatbots, some impediments deter their development, consequently resulting in the low level of customers' value perceptions, satisfaction, and sustained use. Customers express reluctance because chatbots are a new and unfamiliar digital technology (Le, 2023). Customers' interaction and information seeking are also contingent strongly upon offline and online service agents, such as actual employees, social media, and websites (Le, 2023). Chatbot conversations are perceived as unnatural, impersonal, and informal (Jenneboer et al., 2022). Other disadvantages hinder customer experience and engagement, as well, including perceived risk (Abdulquadri et al., 2021; D. M. Nguyen et al., 2021; Nyagadza et al., 2024; Silva et al., 2023), privacy concerns, immature technology (Rese et al., 2020), automation and inconvenience (Nyagadza et al., 2024), intrusiveness (Akdim & Casaló, 2023; Kang & Shao, 2023), lack of awareness of context (Chen et al., 2021), task complexity (X. Cheng et al., 2022) and attachment anxiety (Xu et al., 2022). This leads to customers' resistance and low experience, and consequently, to low usage frequency. In this context, a reasonable explanation for whether or not chatbots dilute customer experience and behavioral outcomes in banking within Vietnam should be put forward.

Despite the growing popularity of chatbots for banking, researchers are concerned about customers' use behaviors. Some prior studies demonstrated how to foster attitude toward chatbots (Cicco et al., 2020), while others focused primarily on enlightening the formation of use intention (Mostafa & Kasamani, 2022; Moussawi et al., 2021; Nyagadza et al., 2024; Rese et al., 2020; Yoon & Yu, 2022), user acceptance (Alnaser et al., 2023), reliance and resistance to chatbots (X. Cheng et al., 2022). However, research on post-adoption intentions toward banking chatbots is insufficient (Ruan & Mezei, 2022). In their work, L. Li et al. (2021) illustrated continuance intention toward chatbots in online travel agencies, with a given concentration on customer confirmation and satisfaction. Similarly, Dhiman and Jamwal (2023) documented continuance intention toward chatbots in tourism due to the role of perceived usefulness and satisfaction. Furthermore, scholars reported that reuse intention toward chatbots is influenced by trust, satisfaction, and perceived usefulness (D. M. Nguyen et al., 2021), language style and interaction (M. Li & Wang, 2023), and flow (Silva et al., 2023). The findings of Park et al. (2022) explained a mechanism of satisfaction and continuance intention in banking chatbots through the impact of emotions, information, and perceived intimacy. Other studies stipulated satisfaction as a critical customer outcome in chatbots (X. Cheng et al., 2024; Eren, 2021; Ruan & Mezei, 2022; Sands et al., 2021; Xu et al., 2022). Researchers like Rahi et al. (2023) stated that a mechanism of continued use of internet banking services must be examined in a contact-less environment. Alnaser et al. (2023) provided insights into the relationship between customer satisfaction and acceptance of AI-enabled banking and the role of AI features (i.e. trendiness, problem solving, and communication quality), perceived performance, and expectation confirmation in boosting customer satisfaction. On the contrary, relatively little is known about the influence of chatbots on customer experience (Sands et al., 2021) and the impact of customer experience on behavioral outcomes in banking (Chen et al., 2021). Therefore, there is room for investigating the variables that matter most in enhancing customer experience and behavioral outcomes in banking chatbots. Moreover, chatbots that desire to optimize customer experience should associate with their distinctive characteristics and quality determinants (Alnaser et al., 2023; Jenneboer et al., 2022; Trivedi, 2019). This study highlights how chatbot attributes induce customer experience in banking. At best, not many studies have paid attention to banking chatbots in an emerging market like that of Vietnam, but the topic has been delved deep into in the world's developed chatbot markets, such as those of the United States (Akdim & Casaló, 2023), China (L. Li et al., 2021; Ruan & Mezei, 2022), India (Dhiman & Jamwal, 2023), Turkey (Eren, 2021) and Lebanon (Mostafa & Kasamani, 2022). To date, recent studies, such as those of Le (2023) and D. M. Nguyen et al. (2021), have clarified behavioral intentions toward chatbots among Vietnamese users, leaving a gap in current literature on customer experience (i.e. extrinsic and intrinsic values) and behavioral outcomes (i.e. satisfaction and continued usage), to which, further empirical work like this is contributed in a developing country.

Together, existing literature has not dwelt much on examining whether or not chatbots are sustainable in banking business communication applications and customer post-adoption behaviors. Additionally, an essential issue that remains unexplained is the understanding of whether or not Vietnamese customers obtain increased satisfaction toward and practice prolonged use of banking chatbots. How chatbot characteristics facilitate and impede customer experience also remains unknown. Building on the aforementioned evidence, this study aims to illuminate customer experience and behavioral outcomes and elucidate the following research questions: (*RQ1*) Does satisfaction boost intention to reuse these chatbots? (*RQ2*) What is the mechanism of customer satisfaction and continuance intention toward banking chatbots? (*RQ3*) Which chatbot attributes affect customer experience?

A research model is tailored to answer these questions. A quantitative study is performed to examine the relationships between chatbot attribute-related factors with customer experience and behavioral outcomes. This work is unique in several aspects. First, this work contributes to bridging the gap in the academic literature by extending the scope of the empirical study on behavioral science and the role of new digital technologies, such as chatbots in banking. Second, it showcases how chatbots leverage customer experience and their influence on customer outcomes toward banking chatbots. Third, following the result presentation, the study provides banks with a useful reference in adopting practical tactics by upgrading important chatbot attribute-related factors determining customer experience and sustained behaviors in an emerging economy.

2. Theoretical background

2.1. Banking chatbots and customer experience

The usage of chatbots for customer support services, that is, for fulfilling customer needs and experience, has substantially increased (Chen et al., 2021). Their usage in banking has aroused growing interest, particularly on chatbots that directly converse with customers for various frontline services (Yoon & Yu, 2022). Chatbots are capable of conveying relevant information, solving deficiencies, streamlining time-consuming tasks, providing prompt responses, and providing customer care support without much human resources. With the absence of employees, they allow banks to deliver services without time constraints, save overhead operating costs, and mitigate impersonal risks related to financial transactions (Abdulquadri et al., 2021; Eren, 2021). Chatbots maximize service effectiveness by substituting for and aiding employees through automated programs, making them crucial for advancing frontline experiences in service encounters (Chen et al., 2021). For these advantages, chatbots are expected to stimulate banks' sustainable development and optimize customer experience.

Customer experience is pondered as aggregate and cumulative customer perception created during learning about, acquiring, using, maintaining, and disposing of a product or service (Carbone & Haeckel, 1994). Customer experience comprises feelings, perceptions, and attitudes while using products or services, which leads to cognitive, emotional, and behavioral responses (Trivedi, 2019). In the digital transformation era, enhancing customer experience is more imperative than ever before (Jenneboer et al., 2022). In the current context, chatbots integrate so well into the customer experience that customers can hardly discern whether these are chatbots or humans they are interacting with (Luo et al., 2019). Customer experience in chatbots attempts to achieve a unique, memorable, and pleasurable experience for customers (Trivedi, 2019). It is viewed as the meaningful way in which customers embrace chatbots for banking. Mathwick et al. (2001) argued that customer experience is measured based on intrinsic and extrinsic values that go well beyond the association of price and quality. Intrinsic values focus on the usage of enjoyment, fun, and pleasure rather than their consequences. On the contrary, extrinsic values concentrate on economic outcomes and are instrumental in nature (Chen et al., 2021). Customer experience from chatbots covers the intrinsic (e.g. enjoyment, fun, and pleasure) (Chen et al., 2021; Trivedi, 2019) and extrinsic values (e.g. efficiency and economic value) (Chen et al., 2021; Jenneboer et al., 2022).

Considering customer experience (i.e. intrinsic and extrinsic values) allows us to elucidate the value perceptions of chatbots, customer satisfaction, and sustained usage toward banking chatbots. The explanation of customers also helps us understand how to improve bank–customer relationships (Le, 2023), enhance customer interactions with all customer-facing value chain members and objects offered by

banks (Trivedi, 2019), and create a long-lasting impression on the customers' thoughts (Chen et al., 2021). The current literature shows a strong impact of customer experience on satisfaction in e-retailing (Chen et al., 2021), brand love in banking (Trivedi, 2019), continued usage and loyalty in shopping (X. Cheng et al., 2022). Following this theoretical position, customers' behavioral outcomes (i.e. satisfaction and continuance intention) will be exceeded by cultivating customer experience toward banking chatbots.

2.2. Information system success (ISS) model

Prior studies sought to investigate the various mechanisms of customer behavioral intentions toward chatbots based on different theories, such as the technology acceptance model (TAM) (Chen et al., 2021; X. Cheng et al., 2024), theory of planned behavior (TPB) (Mostafa & Kasamani, 2022), Search-Access-Test (SAT) (Abdulquadri et al., 2021), unified theory for the acceptance and use of technology (UTAUT) (Nyagadza et al., 2024; Silva et al., 2023) and expectation confirmation model (ECM) (Dhiman & Jamwal, 2023; Eren, 2021; L. Li et al., 2021; D. M. Nguyen et al., 2021). Besides, the Information system success (ISS) is a widely validated model that uncovers the association between the critical characteristics of success in technologies (Okonkwo et al., 2023). The ISS model is prioritized over other models, such as TAM, TPB, SAT, UTAUT, or ECM, because of its effective measurement of critical commercial (i.e. service quality) and technical aspects (i.e. information quality and system quality) (Trivedi, 2019). Following the DeLone and McLean ISS model, the content, service, and system qualities have become core antecedents influencing satisfaction and continuance intention (DeLone & McLean, 2004). Preceding studies utilized this model to gauge customer acceptance of different technologies, such as m-wallet (Okonkwo et al., 2023), massive open online courses (Y.-M. Cheng, 2023) and retail apps (Ng et al., 2024). For conversional agents, the ISS was examined in some settings, for instance, to illuminate customer engagement into chatbots in banking (Trivedi, 2019) and e-retailing (Silva et al., 2023). For banking purposes, earlier studies reflected insights into behavioral intentions toward chatbots. D. M. Nguyen et al. (2021) found that the ISS' quality antecedents are critical requirements of triggering trust, satisfaction, and continuance intention. X. Cheng et al. (2024) unveiled satisfaction and usage intention through the effect of perceived intelligence and virtual identity. Consistent with these arguments, the ISS is identified to suit the scenario of banking chatbots, illuminating the link between quality dimensions, satisfaction, and continuance intention.

In line with a systematic research of Jenneboer et al. (2022) and to better understand the quality dimensions in banking chatbots, we deliberate herein on the several key constructs of chatbot attributes: content quality (D. M. Nguyen et al., 2021; Ruan & Mezei, 2022; Yang & Lee, 2019), system quality, including interaction, competence, and automation (Chung et al., 2020; Jenneboer et al., 2022; Nyagadza et al., 2024; Sands et al., 2021; Xu et al., 2022; Yang & Lee, 2019) and service quality, including personalization, understandability, intimacy, and intrusiveness (Kang & Shao, 2023; L. Li et al., 2021; Ofori et al., 2022; Park et al., 2022; Zhang & Curley, 2018) (see Table 1). Although these attributes motivate customer perceptions (L. Li et al., 2021; D. M. Nguyen et al., 2021; Yang & Lee, 2019), satisfaction (Chen et al., 2021; X. Cheng et al., 2024; Ruan & Mezei, 2022), and continuance intention (Luo et al., 2019; Park et al., 2022; Rese et al., 2020; Sands et al., 2021; Silva et al., 2023), only a few studies collectively examined these relationships in banking, focusing mainly on some salient contexts, such as e-retailing (Chen et al., 2021; Ruan & Mezei, 2022), travelling (L. Li et al., 2021), tourism (Dhiman & Jamwal, 2023), healthcare (Park et al., 2022) and restaurant (Yoon & Yu, 2022). Thus, this study endeavors to enrich the readers' knowledge on behavioral outcomes toward banking chatbots considering the chatbot attributes (see Table 1).

Considering the core recommendations of Chen et al. (2021) and Trivedi (2019), another rather unexplored topic that is ripe for this investigation is the role of chatbot attributes in inducing customer experience in banking and how significant these impacts are (see Table 2). To date, research on the impacts in banking chatbots is rarely conducted (Jenneboer et al., 2022). To extend this stream of research, investigating banking chatbot attributes leading to customer experience is insightful for our work.

3. Hypotheses development and research model

Content quality depicts the customers' perceptions of accuracy, timeliness, and richness of the provision of all content types via electric devices (Jung et al., 2009). Chatbots assist customers by answering questions with different content types, including text, voice, and image. Content quality is appreciated as an

Table 1. Chatbots' quality-related dimension summary.

Studies	Dimensions				
ISS model	Content quality	Service quality	System quality	Customer experience	Other constructs
Trivedi (2019)	Information quality	Service quality	System quality	Customer experience	Brand love and perceived risk
Yang and Lee (2019)	Content quality	x	Automation, visual attractiveness	Extrinsic (usefulness) and intrinsic (enjoyment) values	Behavioral intention
Jenneboer et al. (2022)	Relevance, clarity, valuable information	Assurance, empathy, responsiveness	Response time, reliability, availability, adaptability	Customer experience	Satisfaction, trust, commitment, customer loyalty
Akdim and Casaló (2023)	x	Personalization, intrusiveness	Convenience, compatibility	Perceived value	Continuance intention, recommendation
D. M. Nguyen et al. (2021)	Information quality	Service quality	System quality	x	Satisfaction, continuance intention, trust
Ruan and Mezei (2022)	Information quality	x	Waiting time	Pleasure, arousal	Satisfaction
Xu et al. (2022)	Task-oriented	x	Competence (competent, intelligence, capability, skillfulness)	Warmth perception	Satisfaction
Chen et al. (2021)	x	x	x	Extrinsic and intrinsic values	Satisfaction, usability, responsiveness
Chung et al. (2020)	Communication quality, accuracy, credibility	Customization, Problem solving	Interaction	Entertainment	Satisfaction, trendiness
Rese et al. (2020)	x	x	Convenience, conversation	Intrinsic value (enjoyment and pass time)	Behavioral intention, privacy concerns
Zhang and Curley (2018)	Explanation availability, explanation mode – text	Personalization	Competence	x	Recommendation
Ofori et al. (2022)	x	x	x	Intrinsic (hedonic) value	Satisfaction, continuance intention, economic value
Park et al. (2022)	x	Intimacy	x	Emotional disclosure	Satisfaction, continuance intention
Authors' research model	Accuracy, timeliness, relevance, content related to users' concerns	Personalization, understandability, intimacy, intrusiveness	Interaction, competence, automation	Extrinsic value and intrinsic value	Satisfaction, continuance intention

Table 2. Summary of quality-related constructs in current study.

Constructs	Definition	Sources
Content quality	Perceptions of accuracy, timeliness, richness of the provision of chatbots' content	Trivedi (2019), Yang and Lee (2019), Chung et al. (2020), D. M. Nguyen et al. (2021), Yang and Han (2021), Jenneboer et al. (2022), Ruan and Mezei (2022), Xu et al. (2022)
Personalization	Ability to cognize and treat customers as individuals through personal information and other personal transactions	Zhang and Curley (2018), Chung et al. (2020), Chen et al. (2021), D. M. Nguyen et al. (2021), Akdim and Casaló (2023), Cu (2023)
Understandability	Perceptions that chatbots understand customer commands, a given context of a conversation and the nuance of human language	Chen et al. (2021), L. Li et al. (2021), X. Cheng et al. (2022), Le (2023)
Intimacy	The ambiance of closeness or shared experience of openness	Chen et al. (2021), Song and Shin (2024), X. Cheng et al. (2022), Park et al. (2022)
Intrusiveness	Potential risks when organizations reveal customers' personal information using chatbots	Trivedi (2019), Benlian et al. (2020), Rese et al. (2020), D. M. Nguyen et al. (2021), Nyagadza et al. (2024), Jenneboer et al. (2022), Akdim and Casaló (2023), Kang and Shao (2023), Silva et al. (2023)
Interaction	Perceptions of chatbots' capability to communicate and react to customers' requests and behaviors	Lee and Choi (2017), Chung et al. (2020), Go et al. (2020), Rese et al. (2020), Chen et al. (2021), Sands et al. (2021), Xu et al. (2022), Kang and Shao (2023), Le (2023)
Competence	Perceptions of technology intelligence, knowledge and purpose	Zhang and Curley (2018), Chen et al. (2021), Jenneboer et al. (2022), Xu et al. (2022), Y.-M. Cheng (2023)
Automation	Execution by chatbots in placement of human tasks	Yang and Lee (2019), Jenneboer et al. (2022), X. Cheng et al. (2022), Kang and Shao (2023), Le (2023)

essential criterion for virtual personal assistant technologies (Yang & Lee, 2019). Extant studies demonstrated a strong relationship between content quality and customer behaviors (T. H. Nguyen & Le, 2021). Yang and Lee (2019) found that content quality is a motivation for perceived usefulness and enjoyment of virtual personal assistant devices. Yang and Han (2021) highlighted content quality as a predictor of the extrinsic (utilitarian) and intrinsic (hedonic) values toward the digital technology of virtual reality. D. M. Nguyen et al. (2021) revealed that content quality acts as an important signal for trust and satisfaction because the exact, credible, complete, and up-to-date information provided to customers via chatbots is closely related to their concerns. Ruan and Mezei (2022) indicated that content quality is a determinant of transaction success and satisfaction toward chatbots in the aspect of retailing. Jenneboer et al. (2022) found how content quality (i.e. accuracy, relevance, and value) facilitates customer experience in chatbots. Content quality motivates extrinsic and intrinsic values for customers in chatbots. In this context, dispersing information is necessary for customers when it comes to financial decision-making. If chatbots provide them with inexact, irrelevant, and outdated information, customers fail to reach extrinsic and intrinsic values, consequently leading to poor service experience and dissatisfaction in banks' chatbots. Thus, keeping the preceding discussion in view, we construct the following:

H1a. Content quality positively influences extrinsic value.
H1b. Content quality positively influences intrinsic value.

Interaction reflects customers' perceptions of the technology's capability to communicate and react to their requests and behaviors (Go et al., 2020). In the virtual world, customers desire to seek more information and resolve their issues (Le, 2022a). Through chatbots, customers contact organizations to ask for or receive advice, feel valued, and facilitate transactions (Chung et al., 2020). Accordingly, customers' interaction with organizations becomes easy, and chatbots allow customers to gather information, feel fun, save time, and increase proper decisions (Sands et al., 2021). Le (2023) emphasized that interaction is an indispensable influencing factor for customers' favorable reactions and use behavior toward chatbots. Chung et al. (2020) also affirmed the importance of interaction in communication efficiency, problem resolving, and positive feeling of customers' happiness with chatbots. Kang and Shao (2023) revealed that voice assistants enable two-way communication, improving customers' sense of closeness. Similarly, interaction increased the extrinsic and intrinsic values of customer experience in e-retailing (Chen et al., 2021) and movie recommendation using conversation agent (Lee & Choi, 2017). Thus, it is necessary to improve the bank–customer interaction by using chatbots (Le, 2023). Based on the abovementioned literature, we propose the following hypotheses:

H2a. Interaction positively influences extrinsic value.
H2b. Interaction positively influences intrinsic value.

Competence delineates customers' perceptions of technology intelligence, knowledge, and purpose (Johnson et al., 2008). It is measured by asking customers to rate their perceptions of a robot's intelligence, knowledge, responsibility, and sensibleness (Bartneck et al., 2009). In this study, banking chatbots are perceived as competent because they are intelligent and skillful when promptly responding to customers' requests without employees' intervention. Competence captures perceived intelligence, skillfulness, and capability (Cuddy et al., 2008). Customers depend on competence to infer their experience, intent, and ability (Xu et al., 2022; Zhang & Curley, 2018). Competence is an important driver of customer outcomes (i.e. satisfaction, loyalty, engagement, and well-being) (van Doorn et al., 2017). When customers reach the perceptions of competence of chatbots, such as the ability to complete tasks, provide personalized answers, and understand customers' commands, they are likely to increase the extrinsic value toward chatbots (Y.-M. Cheng, 2023). Chen et al. (2021) demonstrated that chatbots are sensible in identifying a specific context within a conversation, thereby leading to customers' extrinsic and intrinsic values. Jenneboer et al. (2022) asserted that competence is a vital dimension of system quality and motivates customer experience. Chatbots can increase feelings of comfort and help customers achieve personal discussion, address concerns, and receive more useful information about banking services due to their awareness of the customers' given context and understanding of their queries. Based on the abovementioned literature, we propose the following hypotheses:

H3a. Competence positively influences extrinsic value.
H3b. Competence positively influences intrinsic value.

Automation describes the execution by a machine agent of placing human tasks (Parasuraman & Riley, 1997). Automation helps humans enhance affordability and simplicity (Luor et al., 2015). Therefore, automation, which is a sustainable expectation in e-commerce (X. Cheng et al., 2022), e-retailing (Chen et al., 2021), traveling (L. Li et al., 2021) and banking (Eren, 2021), can be realized by AI-based applications (Yang & Lee, 2019). In this study, chatbots are characterized by automation because AI enables them to execute more complicated assignments based on the understanding of customer commands. Consequently, automation boosts perceived usefulness (Yang & Lee, 2019), attitude (Luor et al., 2015) and use behavior (Le, 2023). Kang and Shao (2023) showed that voice assistants, which can independently recommend an implementation scheme to customers, determine themselves ways of conducting tasks, and do things by themselves, will boost customers' psychological ownership, emotions, and well-being. X. Cheng et al. (2022) revealed chatbots' task complexity facilitates customer trust. The above arguments assist us to frame hypotheses:

H4a. Automation positively influences extrinsic value.
H4b. Automation positively influences intrinsic value.

Personalization is the ability of organizations to cognize and treat customers as individuals through personal messaging and other personal transactions (Imhoff et al., 2001). It is an important attribute of recommender agencies with a fundamental influence on use behavior (Zhang & Curley, 2018). Personalization is viewed as a vital service quality dimension that increases trust and satisfaction in bank chatbots (D. M. Nguyen et al., 2021). The identification of customers' queries in a given context of a conversation allows chatbots to elicit customized responses to each customer. Chen et al. (2021) enlightened a mechanism of customer experience based on chatbots' usability. They found that when chatbots possess the capability of conveying specific information and addressing customers' individual needs anywhere and anytime, customers gain extrinsic and intrinsic values. Therefore, personalization is closely related to intrinsic value and favorable assessment (Cu, 2023). Chatbots, with their remarkable ability to provide customers with personalized discussions, will promote better customer experience. Thus:

H5a. Personalization positively influences extrinsic value.
H5b. Personalization positively influences intrinsic value.

Understandability is an important quality criterion for chatbots because it describes customer perceptions that chatbots understand customer commands, given the context of a talk and the nuance of human language (L. Li et al., 2021). It means that chatbots increase the capability of identifying customers' requirements and humor, and then provide appropriate advice and solutions. L. Li et al. (2021) uncovered the relationship between understandability and positive reactions in chatbots. Le (2023) revealed that chatbots improve customer experience when they possess the understanding of customers' dialogues and wants about retailing services. When customers raise awareness of what chatbots listen to and produce better solutions of booking travel packages and flight tickets, they are likely to increase affective statements and expectations. Chen et al. (2021) suggested that understandability is a vital part of chatbot usability. It helps retailers provide complete solutions to customer problems and identify the specific situation during a conversation. Hence, understandability strengthens the extrinsic and intrinsic values toward chatbots in e-retailing. These results confirm the pivotal role of understandability in shaping customer experience toward chatbots, but examining it in banking still remains unclear. Drawing on the preceding opinions, we frame following hypotheses:

H6a. Understandability positively influences extrinsic value.
H6b. Understandability positively influences intrinsic value.

Intimacy depicts the ambiance of closeness or shared experience of openness (Levenson, 1974). Reeves and Nass (1996) explored many dimensions of intimacy in the context of the human–machine link. For example, chatbots are developed to generate feelings of intimacy, such as familiarity (Song & Shin, 2024) and friendliness (X. Cheng et al., 2022). Intimacy has become a crucial dimension of perceived usability and behaviors toward technology (Chen et al., 2021). Intimacy promotes enjoyment (Reis & Shaver, 1988) and trust (X. Cheng et al., 2022). Sung et al. (2007) unveiled a positive impact of intimacy on the pleasure of cleaning and efforts to fit a robot into their homes. As Park et al. (2022) argued, intimacy is a

motivator of customer satisfaction and continuance intention toward chatbots. The current literature supports the influence of intimacy and customer experience and outcomes when communicating with chatbots (Chen et al., 2021). Based on the above discussion, the hypothesis is proposed:

H7. Intimacy positively influences intrinsic value.

To provide more customized services, information is collected and shared by machines to match customer needs (Le & Wang, 2022) and enhance the execution performance (Kang & Shao, 2023). Intrusiveness is alluded as a potential risk when organizations reveal the customers' personal information using chatbots (Silva et al., 2023). Customers will voice their concerns about privacy issues if chatbots control their personal lives, especially under the circumstance of financial services (Nyagadza et al., 2024). Le and Wang (2022) denoted that customers have unfavorable feelings because of the intrusion of personal information and location positioning technology in mobile advertising. In line with this argument, when customers witness private data disclosure or information loss, they would tend to have increased technology anxiety and exhibit reluctance (D. M. Nguyen et al., 2021), dilute trust (Nyagadza et al., 2024) and perceived value toward chatbots (Akdim & Casaló, 2023). Furthermore, intrusiveness from chatbots increases customers' irritation and strain (Benlian et al., 2020), and decreases enjoyment (Rese et al., 2020) and subjective well-being (Kang & Shao, 2023). On the basis of the evidence, the hypothesis is postulated:

H8. Intrusiveness negatively influences intrinsic value.

Extrinsic value reflects the external exposure of a performance through perceived usefulness for gaining value (Davis et al., 1992). We consider the necessity of the extrinsic value for triggering customer outcomes toward chatbots. Le (2022b) enlightened a formation of the extrinsic value and its impact on information-sharing in social media. Researchers have firmly convinced people of the importance of extrinsic values of chatbot agents (i.e. interaction, customization, and problem solving) in promoting customer satisfaction in e-retailing (Chen et al., 2021; Chung et al., 2020) and banking (D. M. Nguyen et al., 2021). Additionally, the current literature shows that extrinsic value is a predictor of continuance intention toward technologies, such as Fintech services (Y. T. H. Nguyen et al., 2022) and social commerce (Hu et al., 2022). Following these arguments, we propose that the effect of the extrinsic value on customer outcomes be considered toward banking chatbots. Grounded in the evident results, following hypotheses are surmised:

H9a. Extrinsic value positively influences satisfaction.
H9b. Extrinsic value positively influences continuance intention.

Meanwhile, intrinsic value portrays customers' feelings of accomplishment, confidence, independence, novelty, and enjoyment toward technology (Venkatesh, 2000). Enjoyment, passing time, pleasure, and arousal are essential intrinsic values and their motivations for attitude and behavioral intentions toward chatbots (Cicco et al., 2020; Moussawi et al., 2021; Yang & Lee, 2019). Customers' enjoyment and hedonic feelings accelerate satisfaction toward technologies (Le, 2022a). Le (2022a) documented that leisure and passing time constitute intrinsic value and are meaningful drivers of satisfaction with social media. Similarly, the intrinsic value is hailed as a determinant of satisfaction in online shopping assistance (Chen et al., 2021; Ruan & Mezei, 2022). Furthermore, it is conceided as a critical facilitator of chatbot use (Moussawi et al., 2021; Silva et al., 2023; Yang & Han, 2021). The existing literature presents a close correlation between intrinsic value and continuance intention toward various technologies, such as location-based advertising (Cu, 2023) and social media (Le, 2022a); however, little is examined in relation to banking chatbots (Eren, 2021). Thus, it is expected that when customers find it fun and enjoyable to make conversations with chatbots, they prolong chatbot usage. Hence, the assumptions are drawn:

H10a. Intrinsic value positively influences satisfaction.
H10b. Intrinsic value positively influences continuance intention.

Satisfaction reflects the psychological states related to customer experience in technology (Oliver, 1980). Satisfaction occurs when customers witness that a chatbot fulfills their expectations and experience in its usage. Consequently, satisfaction boosts sustainable usage and creates long-term customers (Bhattacherjee, 2001 Continuance intention is a post-adoption intention that illustrates customer

intention to continue using technology after adopting initial acceptance (Bhattacherjee, 2001). Scholars affirmed a core association between satisfaction and continuance intention in social media (Le, 2022a) and chatbots (D. M. Nguyen et al., 2021). Moreover, continuance intention is more likely when customers feel satisfied with chatbots in tourism (Dhiman & Jamwal, 2023) and shopping (X. Cheng et al., 2022). Satisfaction after experiencing banks' chatbots is assumed to lead to prolonged usage. Therefore:

H11. Satisfaction positively influences continuance intention.

Figure 1 renders a generic model and hypotheses development.

4. Methodology

4.1. Data collection

A structural questionnaire designed using Google Forms was used to measure the constructs under investigation. Facebook is useful in distributing the questionnaire to the participants because it is one of the most influential social media platforms on Vietnamese people's daily lives, with 66.2 million users in January 2023 (Statista, 2023). Convenience sampling, which describes the process of data collection from the population that is close at hand and easily accessible to researchers (Rahi, 2017), was used as the survey tool. We applied this method because of the absence of a reliable list of participants and its advantages, including cost-time saving and widespread acceptance in behavior science studies (Le, 2022a).

Data were collected in two stages. First, the questionnaire was created based on the adaption of the current literature. We discussed the questionnaire with several researchers in the fields of banking and information system to ensure the validity of all constructs. Thanks to their comments, some major amendments of the items were made, including content quality, automation, intimacy, and intrusiveness. Next, a pilot study was conducted with a small sample (n = 36). The results affirmed that construct validity was satisfactory. The data from the pilot test were excluded in the main survey. Consequently, 39 items of 12 constructs were retained.

Second, the formal data were accumulated. The survey targeted respondents who have used chatbots in Vietnam banks with rising rates of m-payments (D. M. Nguyen et al., 2021). Ethical approval was scrutinized by the research ethics council of Thuongmai University, thereby strengthening adherence to the

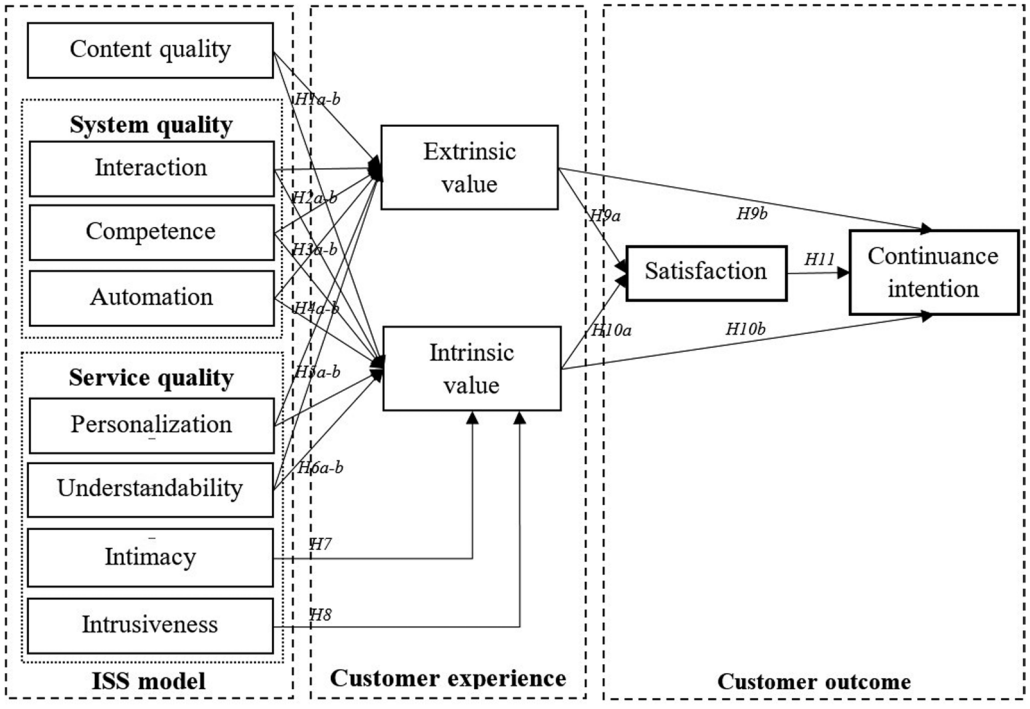


Figure 1. Proposed research model.

ethical standards during the research process. Before filling in the questionnaire, we provided the participants with a written consent form detailing the research purpose, procedures, confidentiality measures, and their rights as respondents. Consent was received in written form because the survey was delivered to the participants for data collection. The participants' participation was completely voluntary. We raised awareness of the right to participate or withdraw from the survey at any time during the research. The anonymity, privacy protection, and confidentiality of the participants' responses were emphasized in the questionnaire's introduction. We provided an introduction of the research purpose and ensured the anonymity and privacy of participants' answers. To enrich the participants' understanding of chatbots, we provided demonstrations (e.g. pictures and videos) before the survey was filled. The demonstrations were non-offensive and understandable and allowed the participants to acquire understanding of AI-based chatbots and their utilization for banking services. A screening question, that is, 'Have you used chatbots in banking before?', in the survey was added to filter the desired sample. The respondents did not receive any incentives for participating in the research. They were encouraged to share the survey link with their peers via social networks to seek potential participants.

According to Hair et al. (2018), the sample size that should be obtained is five times greater (minimum) than the items that desire to be analyzed. The calculation result indicated that 195 respondents is the minimum figure needed. Additionally, Cochran (1977) introduced a sampling formula to calculate the sample size for categorical data:

$$n = Z^2 . \frac{p.(1-p)}{e^2}$$

where: n is the required sample size,Z is the two tailed area under the normal curve where $\alpha = 0.05$ and the Z value is 1.96, e is the acceptable sample error,p is the percentage occurrence of a state or condition (presumed to be 0.5 which maximizes the sample size to be determined).

Given these values and an 'e' value of 6.5%, the sample size is determined as: this draws the acceptable sample size of 267.

However, 361 responses were gathered from July to September 2023. 25 responses were excluded owing to missing and incomplete values. 336 valid responses were yielded for final analysis. Descriptive statistics of respondents are calculated in Table 3.

Table 3 presents respondents' demographics. Most of the respondents were male (57.14%) as compared with few female (42.86%). The majority of the respondents were in the age group of 31 – 40 (37.80%) and 18 – 30 years old (33.33%). Regarding education, over 53.27% had an undergraduate degree, following by 37.80% for graduate and above degree and 8.93% for high school degree. For occupation, 53.57% have had a job, following by student (38.10%), unemployment (6.25%), and other (2.08%). Of the respondents, about 58.63% have utilized banking chatbots for twice in past 3 months, compared to 22.92% for once, 13.99% for thrice, and 4.46% for more than thrice.

Table 3. Respondent demographics.

Demographic		Frequency	Percentage
Gender	Male	192	57.14
	Female	144	42.86
Age	Below 18	23	6.85
	18 – 30	112	33.33
	31 – 40	127	37.80
	> 40	74	22.02
Education	High school	30	8.93
	Undergraduate	179	53.27
	Graduate and above	127	37.80
Occupation	Student	128	38.10
	Working	180	53.57
	Unemployed	21	6.25
	Other	7	2.08
Frequency of using banks' chatbot services in past 3 months	Once	77	22.92
	Twice	197	58.63
	Thrice	47	13.99
	More than thrice	15	4.46

4.2. Measurement

All constructs were adapted from prior validated studies (Le & Nguyen, 2024). Four items measuring content quality from the former study of Jung et al. (2009). Interaction was measured by three items which were adapted from Cho et al. (2019). Three items measuring competence were adapted from Cuddy et al. (2008). Another three items from Zhang and Curley (2018) were used to measure personalization. Automation (four items) was adapted from Luor et al. (2015) and Rijsdijk et al. (2007), understandability (three items) from L. Li et al. (2021) and intimacy (three items) from Berscheid et al. (1989). Three items measuring intrusiveness were adapted from Lau et al. (2019). Three items measuring extrinsic value were adapted from Rose et al. (2012). Three items measuring intrinsic value were adapted from Roy et al. (2018). Similarly, satisfaction was measured by four items which were adapted from Fang et al. (2011), whereas three items measuring continuance intention were adapted from Bhattacherjee (2001).

5. Data analysis and results

5.1. Common method bias

The issue of CMB often occur when all variables are measured with the same respondents at the same time. Harman single factor (HSF) was conducted to CMB on single respondents in the survey. As suggested by Harman (1976), HSF value should be smaller than 50%. The result indicated that HSF value with principal axis factor is 36.227% of the total variance. Therefore, there was no issue on CMB in our data set.

The full collinearity test was used by this study to detect CMB. The outcomes indicated that CMB is not a concern because the variance inflation factor (VIF) values were less than threshold value of 3 (Hair et al., 2018) (Table 4), thereby ensuring data validity.

Furthermore, to avoid CMB, the clarification and transparency of all items of the constructs were provided by conducting pretest. Experts in the fields of banking and information system discussed and suggested several amendments to enhance questionnaire validity. Additionally, the privacy and confidentiality of participants' responses were ensured.

5.2. Measurement model analysis

AMOS 21 was recruited to test the research model. It is a statistical technique with a covariance-based structural equation modeling (CB-SEM). The study used CB-SEM because it is suitable for assessing the fit between the theoretical model and the empirical data. CB-SEM is commonly applied by researchers in marketing and management empirical studies (Steenkamp & Baumgartner, 2000). CB-SEM is covariance-based instead of variance-based (e.g. partial least squares structural equation modeling (PLS-SEM)). Furthermore, it is a statistical approach for examining the relationships using the association between statistical data and qualitative casual hypotheses. CB-SEM aims to test and expand a theory, while PLS-SEM aims to provide predictive applications (Le, 2023). Hypothesis development and assessment of CB-SEM outcomes based on goodness-of-fit criteria are utilized to further emphasize the theory-testing rather than the theory-building character of this methodology (Anderson & Gerbing, 1988).

The study evaluated the measurement model based on convergent and discriminant validity. For the convergent validity, we recruited two criteria, namely, composite reliability (CR) and average variance

Table 4. Multi-collinearity test.

	VIF
Content quality	1.393
Interaction	1.842
Competence	1.701
Automation	1.286
Personalization	1.948
Understandability	1.990
Intimacy	1.253
Intrusiveness	1.122

extracted (AVE). These criteria should be greater than 0.70 (for CR) and 0.50 (for AVE) (Hair et al., 2018). The results showed that all the CR and AVE values exceeded 0.70 and 0.50, respectively, suggesting the convergent validity for the measures.

The reliability of the measures was determined based on Cronbach's alpha (CA),which should be greater than 0.70 (Hair et al., 2018). The results indicated that the CA values of all the variables met the abovementioned recommended threshold (Table 5), suggesting the good reliability of the measures.

To assess the discriminant validity, the square root of the AVE for each construct must be higher than all the correlation coefficients with other constructs (Fornell & Larcker, 1981). The diagonal constructs, which are in bold in the correlation matrix, are the square root of the AVE, while the off-diagonal constructs are the simple correlation coefficient between the corresponding constructs. The results presented that all square roots of AVE values are greater than all coefficient correlations (Table 6).

Additionally, the study selected several criteria to gauge the goodness of the structural model: the ratio between the Chi square and the degree of freedom (χ^2/df), TFI, CFI, NFI, IFI, and RMSEA. In accordance with Hair et al. (2018), χ^2/df and RMSEA should be smaller than 3 and 0.08, respectively, while the TFI, CFI, NFI, and IFI should be higher than 0.9. The results confirmed a good model fit for the model (Table 7).

5.3. Structural model analysis

Results of hypothesized relationships are shown in Table 8 and Figure 2. The study supports 18 hypotheses. This study explains 46.9%, 52.1%, 21.8% and 63% of variation of extrinsic value, intrinsic value, satisfaction and continuance intention toward banking chatbots respectively.

First, the content quality ($\beta = 0.131$; $p = 0.019$), interaction ($\beta = 0.138$; $p = 0.021$), competence ($\beta = 0.161$; $p = 0.019$), automation ($\beta = 0.166$; $p = 0.010$), personalization ($\beta = 0.137$; $p = 0.024$), and understandability ($\beta = 0.175$; $p = 0.022$) positively affect the extrinsic value. Therefore, hypotheses H1a – H6a are supported.

Second, the content quality ($\beta = 0.123$; $p = 0.024$), interaction ($\beta = 0.128$; $p = 0.030$), competence ($\beta = 0.175$; $p = 0.022$), personalization ($\beta = 0.179$; $p = 0.003$), understandability ($\beta = 0.207$; $p = 0.005$), and intimacy ($\beta = 0.134$; $p = 0.009$) positively affect the intrinsic value, thereby supporting H1b – H3b, H5b, and H6b. Additionally, intrusiveness ($\beta = -0.107$; $p = 0.019$) negatively affects the intrinsic value, thereby supporting H8. However, H4b is not supported because automation ($\beta = 0.088$; $p = 0.165$) does not influence the intrinsic value.

Third, H9a and H9b are supported when the extrinsic value positively affects satisfaction ($\beta = 0.188$; $p < 0.001$) and continuance intention ($\beta = 0.305$; $p < 0.001$). Likewise, the intrinsic value has a positive effect on satisfaction ($\beta = 0.182$; $p < 0.001$) and continuance intention ($\beta = 0.377$; $p < 0.001$); therefore, H10a – H10b are supported. Lastly, H11 is supported based on the positive effect of satisfaction ($\beta = 0.523$; $p < 0.001$) on intention to continue using banking chatbots.

6. Discussion and implications

6.1. Theoretical implications

Several theoretical implications are drawn. Our study enriches the understanding of customer outcomes of post-adoption intention toward AI-based chatbots in banking, while the majority of past studies investigated the adoption intention toward chatbots in retailing (Cicco et al., 2020; Mostafa & Kasamani, 2022; Rese et al., 2020; Silva et al., 2023), travelling and tourism (Dhiman & Jamwal, 2023; L. Li et al., 2021), healthcare (Park et al., 2022) and restaurant (Yoon & Yu, 2022). Additionally, scholars revealed different mechanisms of customer satisfaction toward chatbots (Chen et al., 2021; Eren, 2021; Ruan & Mezei, 2022), but did not thoughtfully examine the consequences of satisfaction. Accordingly, the current study responds to their call and extends insights into the relationship between satisfaction and post-adoption intention (i.e. continuance intention). This study also provides support to prior studies as regards the formation of satisfaction with banking chatbots. The influence of the extrinsic and intrinsic values on satisfaction is investigated, which is consistent with that in the study of Chen et al. (2021) and D. M. Nguyen et al. (2021). The findings hint on the following: the greater the customer experience with

Table 5. Reliability and validity of the measurement.

Construct	Std-loadings	CA	AVE	CR
Content quality				
COQ1. Content provided by chatbots is accurate	0.890	0.940	0.797	0.940
COQ2. Content provided by chatbots is sufficiently timely	0.885			
COQ3. Content provided by chatbots is relevant to my decision-making	0.903			
COQ4. Chatbots provide content pertaining to my concerns	0.894			
Interaction		0.896	0.745	0.897
ACT1. I am in control of my personal needs through chatbots	0.839			
ACT2. I perceive chatbots to be sensitive to my personal needs	0.846			
ACT3. Chatbots provide opportunities to give my responses	0.903			
Competence		0.904	0.758	0.904
COM1. Chatbots are competent	0.868			
COM2. Chatbots are intelligent	0.872			
COM3. Chatbots are skillful	0.872			
Automation		0.887	0.668	0.889
AUT1. It is convenient that chatbots help users proactively without human intervention	0.860			
AUT2. It is convenient that chatbots provide auto-adjusted control	0.819			
AUT3. Chatbots autonomously provide me the choice of what to do	0.823			
AUT4. Chatbots independently provide me recommendations for action plans for assigned matters	0.764			
Personalization		0.957	0.881	0.957
PER1. Chatbots understand my needs	0.930			
PER2. Chatbots know what I want	0.938			
PER3. The advice appears to tailored for me personally	0.948			
Understandability		0.907	0.765	0.907
UND1. I feel that what I am saying to chatbots is well understood by the system	0.904			
UND2. I feel that the words in my questions are well understood by chatbots	0.835			
UND3. I feel that chatbots understand my intentions when I ask questions to it	0.883			
Intimacy		0.920	0.793	0.920
INT1. I develop a sense of familiarity with chatbots	0.905			
INT2. Chatbots use supportive statements to build favor with me	0.870			
INT3. I feel emotionally close to chatbots	0.896			
Intrusiveness		0.925	0.807	0.926
TRU1. While receiving responses from chatbots, I feel I am under surveillance	0.887			
TRU2. While receiving responses from chatbots, I feel I am being monitored	0.910			
TRU3. While receiving responses from chatbots, I feel they are listening to everything around me	0.897			
Extrinsic value		0.916	0.785	0.916
EVA1. Chatbots make me feel that they are talking to me personally as a customer	0.873			
EVA2. Chatbots help resolve my needs without other problems	0.902			
EVA3. Chatbots make me feel valued as a customer	0.883			
Intrinsic value		0.912	0.780	0.914

(*Continued*)

Table 5. Continued.

Construct	Std-loadings	CA	AVE	CR
IVA1. I like chatbots when they help me customize my financial experience to my own liking	0.890			
IVA2. I enjoy getting the benefits from using chatbots with little effort	0.835			
IVA3. Chatbots are fun to converse with	0.923			
Satisfaction		0.884	0.658	0.885
SAT1. I am pleased with using chatbots	0.781			
SAT2. I like to use chatbots from the bank websites	0.802			
SAT3. I think that using chatbots on the bank website is a good idea	0.854			
SAT4. Overall, I am satisfied with the experience of using chatbots	0.805			
Continuance intention		0.913	0.782	0.915
COI1. My intention is to continue using chatbots over other alternative means of communication or searching tools on the bank websites	0.835			
COI2. All things considered, I expect to continue using chatbots often in the future	0.905			
COI3. I can see myself increasing the use of chatbots if possible	0.911			

Table 6. Discriminant validity.

	COQ	ACT	COM	AUT	PER	UND	INT	TRU	EVA	IVA	SAT	COI
COQ	**0.893**											
ACT	0.396	**0.863**										
COM	0.484	0.557	**0.871**									
AUT	0.255	0.403	0.355	**0.817**								
PER	0.457	0.599	0.537	0.409	**0.939**							
UND	0.460	0.632	0.591	0.378	0.660	**0.875**						
INT	0.280	0.409	0.367	0.346	0.322	0.348	**0.891**					
TRU	0.246	0.244	0.280	0.231	0.272	0.259	0.164	**0.898**				
EVA	0.439	0.543	0.523	0.414	0.322	0.565	0.322	0.249	**0.886**			
IVA	0.444	0.564	0.534	0.386	0.549	0.595	0.414	0.133	0.473	**0.883**		
SAT	0.554	0.361	0.416	0.255	0.459	0.461	0.226	0.144	0.381	0.376	**0.811**	
COI	0.510	0.571	0.654	0.368	0.597	0.623	0.340	0.253	0.591	0.631	0.613	**0.884**

Table 7. Model fit.

Index	Threshold value	Structural model
χ^2/df	<3.0	0.149
TLI	≤0.9	0.967
CFI	≤0.9	0.971
NFI	≤0.9	0.917
IFI	≤0.9	0.971
RMSEA	<0.08	0.038

extrinsic and intrinsic values toward chatbots, the greater the satisfaction with these chatbots. Our study is among the firsts to illuminate the role of AI-based chatbots in satisfaction and continuance intention in banking.

Second, the study contributes to the ISS model in the setting of AI-based chatbots and banking, while preceding studies focused on several grounded theories to enlighten satisfaction and behavioral intentions toward chatbots, such as the expectation–confirmation model (L. Li et al., 2021), perspective of the product attribute type (Ruan & Mezei, 2022), social actor paradigm (Park et al., 2022), TAM (X. Cheng et al., 2024) and task-technology fit model (Dhiman & Jamwal, 2023). In other words, this study addresses

Table 8. Hypothesis testing.

Hypothesis	Path	Coefficient	*p*-value	Decision
H1a	COQ→ EVA	0.131*	0.019	Supported
H1b	COQ→ IVA	0.123*	0.024	Supported
H2a	ACT→EVA	0.138*	0.021	Supported
H2b	ACT→IVA	0.128*	0.030	Supported
H3a	COM→ EVA	0.161*	0.019	Supported
H3b	COM→ IVA	0.175*	0.022	Supported
H4a	AUT→ EVA	0.166**	0.010	Supported
H4b	AUT→ IVA	0.088$^{n/s}$	0.165	Not supported
H5a	PER→EVA	0.137*	0.024	Supported
H5b	PER→IVA	0.179**	0.003	Supported
H6a	UND→ EVA	0.175*	0.022	Supported
H6b	UND→ IVA	0.207**	0.005	Supported
H7	INT→ IVA	0.134**	0.009	Supported
H8	TRU→ IVA	−0.107*	0.019	Supported
H9a	EVA→SAT	0.188***	0.000	Supported
H9b	EVA→COI	0.305***	0.000	Supported
H10a	IVA→SAT	0.182***	0.000	Supported
H10b	IVA→COI	0.377***	0.000	Supported
H11	SAT→COI	0.523***	0.000	Supported

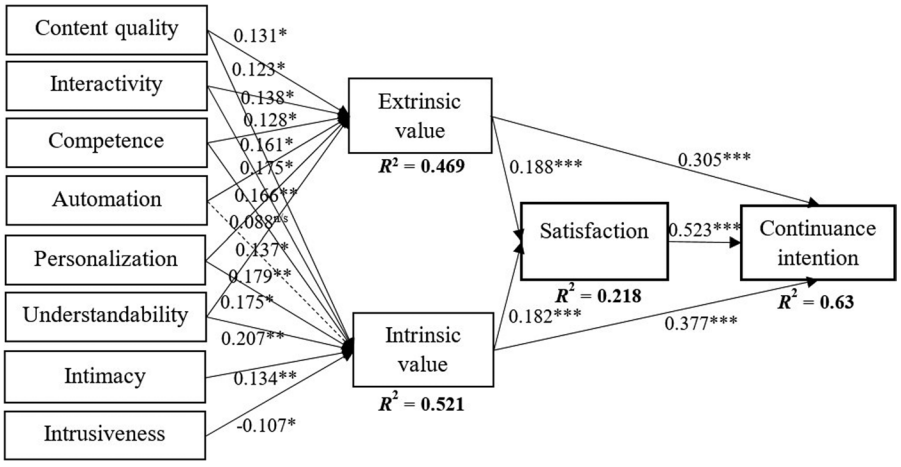


Figure 2. The structural model.

the research gap by identifying the antecedents that matter most in leveraging customer experience and behavioral outcomes in chatbots.

Third, the results extend knowledge by revealing how chatbot attributes promote extrinsic and intrinsic values in banking. We attempted to develop herein the research framework of Chen et al. (2021) asserting the impact of chatbot characteristics on customer experience in e-retailing. Nonetheless, their study acknowledged that, besides usability and responsiveness, other chatbot attributes should be involved to foster customer experience. Taking this necessity into consideration, the current study addresses this limitation and extends the research frameworks of prior studies that focused on how chatbots drive customer experience by adapting crucial chatbot attributes, namely, content quality, interaction, and understandability. These findings are in tandem with the affirmations of previous studies (L. Li et al., 2021; Ruan & Mezei, 2022). When customers perceive chatbots as a fertile source of valuable information and tend to consider them as being somewhat helpful when having conversations with and in understanding their intentions, they would have a more interesting experience. The results also reveal that the role of other attributes, namely, competence, automation, personalization, and intimacy, in heightening the intrinsic and extrinsic values is different from that in preliminary studies, which enriched the ISS model. Researchers endeavored to explicate behavioral intention through the effects of competence (X. Cheng et al., 2024; Moussawi et al., 2021; Zhang & Curley, 2018), automation (Nyagadza et al., 2024; Yang & Lee, 2019), personalization (Zhang & Curley, 2018) and intimacy (Park et al., 2022), while there is a lack of investigation on the influence of these constructs on customer experience in banking chatbots (Jenneboer et al., 2022). Additionally, the negative link between intrusiveness and intrinsic value

is uncovered in the banking chatbot scenario, hinting that customers express unfavorable emotions toward chatbot invasion. Chatbots are intrusive when they can interfere with customers' lives by gathering personal information (Kang & Shao, 2023). Collectively, this study provides a holistic explanation on the influence of chatbot attribute-related factors on customer experience.

6.2. Practical implications

Some practical implications are brought for practitioners. Overall, it is indispensable for banks to encourage customers' continued usage of chatbots by generating opportunities that help customers understand technological trends and gain experience related to chatbots. In the commencement of chatbots, marketers must focus much effort on developing practical solutions that induce customers' sustainable usage of chatbots for banking services. The findings denote that satisfaction and continuance intention are accelerated by customer experience. Practitioners should understand how chatbots are designed to raise customer values and post-adoption behaviors.

First, given the role played by content quality, practitioners must develop chatbots that generate complete, accurate, relevant, and up-to-date information, as AI-based chatbots tend to be user-centered. Timely answers are available for use as an instant response to customers' queries. Second, interaction appears to be a motivator of customer experience. Banks must improve communication with customers using chatbots. Marketers must stress the relative advantages of chatbots' seamless reciprocity in real time without delay and errors. Marketers must also bring the simplicity of chatbot utilization via customized interactions. Available responses (24/7) from chatbots help customers converse with banks at a low cost. When chatbots are still in their embryonic stage, other interactive tools should be equipped in parallel with chatbots to raise the banking service quality. This could be online (e.g. social networks) and offline (e.g. staffs). Third, a significant association between competence and customer experience suggests that banks should enhance chatbots' intelligence by precisely predicting customers' expectations, promptly receiving their requests, quickly responding to their questions without averseness, learning from customers' new information, readily adapting changes, and successfully accomplishing tasks within a timeframe for customers. Fourth, to ensure chatbots' performance in increasing automation, practitioners must improve assignment execution by chatbots without human intervention and minimize the inexact responses because these failures do not fulfill customers' expectations on information searching and mitigate trust, emotions, and satisfaction in banking sustainability. Furthermore, for a mechanism of customer experience in chatbots, banks should pay attention to the other distinctive features of banks' chatbots.

Fifth, considering personalization as a facilitator of extrinsic and intrinsic values, banks attempt to deliver personalized messages and resolve issues about banking services based on the chatbots' ability to analyze the customers' given situations, foresee potential queries, and make conversations using the customers' personal information (e.g. names) and natural speech (e.g. greetings and valedictions). If these messages are not closely combined with customer preferences, chatbots' usefulness and customers' affective states can be reduced (Le, 2023). Sixth, the necessity to strengthen customer values means to concentrate on understandability. Banks considerably strive hard to identify market segments, define customer demands, and prepare proper scripts. In parallel with the elaborate preparation of chatbot programs, banks must collect customers' comments on chatbots' effectiveness through interviews and surveys. Consequently, banks would obtain customer insights and satisfy their demands. Seventh, when customers feel as friendly to chat with chatbots, they would raise their intrinsic value. Banks try to exhibit warmth and show concern for customers by using convincing language with natural and sentimental statements. Practitioners develop chatbots' voice features and alter their language styles to increase familiarity and interpret customers' instructions. Thanks to the sense of anthropomorphism, customers trigger positive emotions. Lastly, the finding indicating the effect of intrusiveness on the intrinsic value highlights a strong recommendation. The enormous hurdle of intrusiveness should be adequately solved. Banks must develop chatbots' safe technology methods, adopt policies and principles, and find appropriate methods of collecting customer information, therefore tackling the growing concerns about customers' sense of surveillance and worries about the security of banking chatbots (Cicco et al., 2020; D. M. Nguyen et al., 2021).

7. Conclusion and future research directions

The study presents an insightful contribution to the current literature on consumer sustainable behaviors toward banking chatbots. Although AI-based chatbots are pivotal for banks to improve reciprocity with customers, little understanding is unveiled about how banking chatbots induce customer experience and sustainable behaviors. To respond timely to the call, a continuance intention model was extended herein based on the ISS success model associated with chatbots' salient attributes. The results affirmed that customers enjoy the new experience of using banking services because of the assistance of AI-based chatbots. Furthermore, customer experiences were investigated to evoke satisfaction and intention to continue banking chatbots usage. Our findings assist banks in enriching the understanding of customer post-adoption behaviors and adopting practical solutions for enhancing satisfaction and prolonged usage of chatbots in Vietnam.

Several limitations must be noted. First, data for our study, which were accumulated from a sample size of 336 respondents, may lead to a limited generalizability of the findings to the population. Therefore, a larger sample with diverse respondents should be employed by future studies to increase the generalization and offer more significant contributions. Second, continuance intention was alluded as an essential behavioral outcome in our study, but customers may adopt other post-adoption behaviors, such as recommendation and information seeking (Jenneboer et al., 2022; Zhang & Curley, 2018). Thus, the abovementioned intentions may be encompassed for further studies to gain deep insights into customers' sustained behaviors toward banking chatbots. Third, our model was empirically examined in a specific banking context, while AI-based chatbots are utilized in various fields, such as retailing, tourism, healthcare, and restaurant. Interestingly, a comparative study must be conducted among these fields under various circumstances. Fourth, the shortcoming of the current study is the performed cross-sectional study, which led to the limitation of explaining customer experience and prolonged behaviors over time. Promising studies should focus on a longitudinal research to offer holistic insights into customer experience and continuance intention toward chatbots. Fifth, although the value of continuance intention (R^2 = 63%) was shown by the results, complementary constructs, such as trust, usability, and emotional responses, may be involved in this behavior. Thus, a comprehensive model could be tailored using additional constructs in the next research.

Author contributions

Conceptualization: Tran Hung Nguyen and Xuan Cu Le; design: Tran Hung Nguyen; acquisition: Tran Hung Nguyen and Xuan Cu Le; analysis: Xuan Cu Le; interpretation of the data: Tran Hung Nguyen; the draft of the paper: Xuan Cu Le; revising it critically for intellectual content: Tran Hung Nguyen and Xuan Cu Le. The final approval of the version to be published, and that all authors agree to be accountable for all aspects of the work.

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Data availability statement

The data present in this study are available on request from the corresponding author.

References

Abdulquadri, A., Mogaji, E., Kieu, T. A., & Nguyen, N. P. (2021). Digital transformation in financial services provision: A Nigerian perspective to the adoption of chatbot. *Journal of Enterprising Communities: People and Places in the Global Economy*, *15*(2), 258–281. https://doi.org/10.1108/JEC-06-2020-0126

Akdim, K., & Casaló, L. V. (2023). Perceived value of AI-based recommendations service: The case of voice assistants. *Service Business*, *17*(1), 81–112. https://doi.org/10.1007/s11628-023-00527-x

Alnaser, F. M., Rahi, S., Alghizzawi, M., & Ngah, A. H. (2023). Does artificial intelligence (AI) boost digital banking user satisfaction? Integration of expectation confirmation model and antecedents of artificial intelligence enabled digital banking. *Heliyon*, *9*(8), e18930. https://doi.org/10.1016/j.heliyon.2023.e18930

Anderson, J. C., & Gerbing, D. (1988). Structural equation modeling in practice: A review of recommended two-step approach. *Psychological Bulletin*, *103*(3), 411–423. https://doi.org/10.1037/0033-2909.103.3.411

Bartneck, C., Kulić, D., Croft, E., & Zoghbi, S. (2009). Measurement instruments for the anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety of robots. *International Journal of Social Robotics*, *1*(1), 71–81. https://doi.org/10.1007/s12369-008-0001-3

Benlian, A., Klumpe, J., & Hinz, O. (2020). Mitigating the intrusive effects of smart home assistants by using anthropomorphic design features: A multimethod investigation. *Information Systems Journal*, *30*(6), 1010–1042. https://doi.org/10.1111/isj.12243

Berscheid, E., Snyder, M., & Omoto, A. M. (1989). The relationship closeness inventory: Assessing the closeness of interpersonal relationships. *Journal of Personality and Social Psychology*, *57*(5), 792–807. https://doi.org/10.1037/0022-3514.57.5.792

Bhattacherjee, A. (2001). Understanding information systems continuance: An expectation-confirmation model. *MIS Quarterly*, *25*(3), 351–370. https://doi.org/10.2307/3250921

Carbone, L., & Haeckel, S. (1994). Engineering customer experiences. *Marketing Management*, *3*(3), 1–9.

Chen, J.-S., Le, T.-T.-Y., & Florence, D. (2021). Usability and responsiveness of artificial intelligence chatbot on online customer experience in e-retailing. *International Journal of Retail & Distribution Management*, *49*(11), 1512–1531. https://doi.org/10.1108/IJRDM-08-2020-0312

Cheng, Y.-M. (2023). Which quality determinants cause MOOCs continuance intention? A hybrid extending the expectation-confirmation model with learning engagement and information systems success. *Library Hi Tech*, *41*(6), 1748–1780. https://doi.org/10.1108/LHT-11-2021-0391

Cheng, X., Bao, Y., Zarifis, A., Gong, W., & Mou, J. (2022). Exploring consumers' response to text-based chatbots in e-commerce: The moderating role of task complexity and chatbot disclosure. *Internet Research*, *32*(2), 496–517. https://doi.org/10.1108/INTR-08-2020-0460

Cheng, X., Qiao, L., Yang, B., & Li, Z. (2024). An investigation on the influencing factors of elderly people's intention to use financial AI customer service. *Internet Research*, *34*(3), 690–717. https://doi.org/10.1108/INTR-06-2022-0402

Cho, W.-C., Lee, K. Y., & Yang, S.-B. (2019). What makes you feel attached to smartwatches? The stimulus–organism–response (S–O–R) perspectives. *Information Technology & People*, *32*(2), 319–343. https://doi.org/10.1108/ITP-05-2017-0152

Chung, M., Ko, E., Joung, H., & Kim, S. J. (2020). Chatbot e-service and customer satisfaction regarding luxury brands. *Journal of Business Research*, *117*, 587–595. https://doi.org/10.1016/j.jbusres.2018.10.004

Cicco, R. D., Silva, S. C., & Alparone, F. R. (2020). Millennials' attitude toward chatbots: An experimental study in a social relationship perspective. *International Journal of Retail & Distribution Management*, *48*(11), 1213–1233. https://doi.org/10.1108/IJRDM-12-2019-0406

Cochran, W. G. (1977). *Sampling techniques* (3rd ed.). Wiley & Sons.

Cu, L. X. (2023). A hedonic value-based consumer continuance intention model toward location-based advertising. *Revista de Gestão*, *31*(1), 34–49. https://doi.org/10.1108/REGE-08-2021-0165

Cuddy, A. J. C., Fiske, S. T., & Glick, P. (2008). Warmth and competence as universal dimensions of social perception: The stereotype content model and the BIAS map. In *Advances in experimental social psychology* (Vol. 40, pp. 61–149). Academic Press. https://doi.org/10.1016/S0065-2601(07)00002-0

Davis, F. D., Bagozzi, R. P., & Warshaw, P. R. (1992). Extrinsic and intrinsic motivation to use computers in the workplace. *Journal of Applied Social Psychology*, *22*(14), 1111–1132. https://doi.org/10.1111/j.1559-1816.1992.tb00945.x

DeLone, W. H., & McLean, E. R. (2004). Measuring e-commerce success: Applying the DeLone & McLean information systems success model. *International Journal of Electronic Commerce*, *9*(1), 31–47. http://www.jstor.org/stable/27751130 https://doi.org/10.1080/10864415.2004.11044317

Dhiman, N., & Jamwal, M. (2023). Tourists' post-adoption continuance intentions of chatbots: Integrating task–technology fit model and expectation–confirmation theory. *foresight*, *25*(2), 209–224. https://doi.org/10.1108/FS-10-2021-0207

Eren, B. A. (2021). Determinants of customer satisfaction in chatbot use: Evidence from a banking application in Turkey. *International Journal of Bank Marketing*, *39*(2), 294–311. https://doi.org/10.1108/IJBM-02-2020-0056

Fang, Y. H., Chiu, C. M., & Wang, E. T. G. (2011). Understanding customers' satisfaction and repurchase intentions. *Internet Research*, *21*(4), 479–503. https://doi.org/10.1108/10662241111158335

Fornell, C., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research*, *18*(1), 39–50. https://doi.org/10.2307/3151312

Go, H., Kang, M., & Suh, S. C. (2020). Machine learning of robots in tourism and hospitality: Interactive technology acceptance model (iTAM) – cutting edge. *Tourism Review*, *75*(4), 625–636. https://doi.org/10.1108/TR-02-2019-0062

Ha, M. S., & Nguyen, T. L. (2022). Digital transformation in banking: A case from Vietnam. In T. Phan & D. Damian (Eds.), *Smart cities in Asia: regulations, problems, and development* (pp. 103–114). Springer Nature Singapore. https://doi.org/10.1007/978-981-19-1701-1_9

Hair, F., Babin, B., Black, W. J., & Anderson, R. E. (2018). *Multivariate data analysis*. Cengage.

Harman, H. H. (1976). *Modern factor analysis*. University of Chicago Press.

Hu, X., Chen, Z., Davison, R. M., & Liu, Y. (2022). Charting consumers' continued social commerce intention. *Internet Research*, *32*(1), 120–149. https://doi.org/10.1108/INTR-07-2020-0397

Huong, L. N. Q., & Binh, N. H. (2022). Chatbot trong linh vuc ngan hang – Thuc trang va xu huong ung dung tai Viet Nam. *Journal of Economic & Banking*, *Studies*(236 + 237), 64–78.

Imhoff, C., Loftis, L., & Geiger, J. (2001).). *Building the customer-centric enterprise, data warehousing techniques for supporting customer relationship management*. Wiley.

Jenneboer, L., Herrando, C., & Constantinides, E. (2022). The Impact of chatbots on customer loyalty: A systematic literature review. *Journal of Theoretical and Applied Electronic Commerce Research*, *17*(1), 212–229. https://doi.org/10.3390/jtaer17010011

Johnson, R. D., Marakas, G. M., & Palmer, J. W. (2008). Beliefs about the social roles and capabilities of computing technology: Development of the computing technology continuum of perspective. *Behaviour & Information Technology*, *27*(2), 169–181. https://doi.org/10.1080/01449290600959104

Jung, Y., Perez-Mira, B., & Wiley-Patton, S. (2009). Consumer adoption of mobile TV: Examining psychological flow and media content. *Computers in Human Behavior*, *25*(1), 123–129. https://doi.org/10.1016/j.chb.2008.07.011

Kang, W., & Shao, B. (2023). The impact of voice assistants' intelligent attributes on consumer well-being: Findings from PLS-SEM and fsQCA. *Journal of Retailing and Consumer Services*, *70*, 103130. https://doi.org/10.1016/j.jretconser.2022.103130

Lau, C. K. H., Chui, C. F. R., & Au, N. (2019). Examination of the adoption of augmented reality: A VAM approach. *Asia Pacific Journal of Tourism Research*, *24*(10), 1005–1020. https://doi.org/10.1080/10941665.2019.1655076

Le, X. C. (2022a). Charting sustained usage toward mobile social media application: The criticality of expected benefits and emotional motivations. *Asia Pacific Journal of Marketing and Logistics*, *34*(3), 576–593. https://doi.org/10.1108/APJML-11-2020-0779

Le, X. C. (2022b). Propagation of information-sharing in social media: The perspective of intrinsic and extrinsic cues. *VINE Journal of Information and Knowledge Management Systems*, *54*(5), 973–989. https://doi.org/10.1108/VJIKMS-01-2022-0006

Le, X. C. (2023). Inducing AI-powered chatbot use for customer purchase: The role of information value and innovative technology. *Journal of Systems and Information Technology*, *25*(2), 219–241. https://doi.org/10.1108/JSIT-09-2021-0206

Lee, S., & Choi, J. (2017). Enhancing user experience with conversational agent for movie recommendation: Effects of self-disclosure and reciprocity. *International Journal of Human-Computer Studies*, *103*, 95–105. https://doi.org/10.1016/j.ijhcs.2017.02.005

Le, X. C., & Nguyen, T. H. (2024). The effects of chatbot characteristics and customer experience on satisfaction and continuance intention toward banking chatbots: Data from Vietnam. *Data in Brief*, *52*, 110025. https://doi.org/10.1016/j.dib.2023.110025

Levenson, E. A. (1974). Changing concepts of intimacy in psychoanalytic practice. *Contemporary Psychoanalysis*, *10*(3), 359–369. https://doi.org/10.1080/00107530.1974.10745341

Le, X. C., & Wang, H. (2022). Context-aware and social integrative-related factors as the precursors of efficient context aware advertising via mobile applications. *International Journal of Mobile Communications*, *20*(3), 332–348. https://doi.org/10.1504/IJMC.2022.10035486

Li, L., Lee, K. Y., Emokpae, E., & Yang, S.-B. (2021). What makes you continuously use chatbot services? Evidence from Chinese online travel agencies. *Electronic Markets*, *31*(3), 575–599. https://doi.org/10.1007/s12525-020-00454-z

Li, M., & Wang, R. (2023). Chatbots in e-commerce: The effect of chatbot language style on customers' continuance usage intention and attitude toward brand. *Journal of Retailing and Consumer Services*, *71*, 103209. https://doi.org/10.1016/j.jretconser.2022.103209

Luo, X., Tong, S., Fang, Z., & Qu, Z. (2019). Frontiers: Machines vs. humans: The impact of artificial intelligence chatbot disclosure on customer purchases. *Marketing Science*, *38*(6), 937–947. https://doi.org/10.1287/mksc.2019.1192

Luor, T., Lu, H.-P., Yu, H., & Lu, Y. (2015). Exploring the critical quality attributes and models of smart homes. *Maturitas*, *82*(4), 377–386. https://doi.org/10.1016/j.maturitas.2015.07.025

Mathwick, C., Malhotra, N., & Rigdon, E. (2001). Experiential value: Conceptualization, measurement and application in the catalog and Internet shopping environment. *Journal of Retailing*, *77*(1), 39–56. https://doi.org/10.1016/S0022-4359(00)00045-2

Mostafa, R. B., & Kasamani, T. (2022). Antecedents and consequences of chatbot initial trust. *European Journal of Marketing*, *56*(6), 1748–1771. https://doi.org/10.1108/EJM-02-2020-0084

Moussawi, S., Koufaris, M., & Benbunan-Fich, R. (2021). How perceptions of intelligence and anthropomorphism affect adoption of personal intelligent agents. *Electronic Markets*, *31*(2), 343–364. https://doi.org/10.1007/s12525-020-00411-w

Ng, S. L., Rezaei, S., Valaei, N., & Iranmanesh, M. (2024). Modelling services continuance intention: Evidence from apps stores. *Asia-Pacific Journal of Business Administration*, *16*(2), 256–281. https://doi.org/10.1108/APJBA-08-2021-0408

Nguyen, D. M., Chiu, Y.-T. H., & Le, H. D. (2021). Determinants of continuance intention towards banks' chatbot services in Vietnam: A necessity for sustainable development. *Sustainability*, *13*(14), 7625. https://doi.org/10.3390/su13147625

Nguyen, T. H., & Le, X. C. (2021). How social media fosters the elders' COVID-19 preventive behaviors: Perspectives of information value and perceived threat. *Library Hi Tech*, *39*(3), 776–795. https://doi.org/10.1108/LHT-09-2020-0241

Nguyen, T. H., Le, X. C., & Vu, T. H. L. (2022). An extended technology-organization-environment (TOE) framework for online retailing utilization in digital transformation: Empirical evidence from Vietnam. *Journal of Open Innovation: Technology, Market, and Complexity*, *8*(4), 200. https://www.mdpi.com/2199-8531/8/4/200 https://doi.org/10.3390/joitmc8040200

Nguyen, Y. T. H., Tapanainen, T., & Nguyen, H. T. T. (2022). Reputation and its consequences in Fintech services: The case of mobile banking. *International Journal of Bank Marketing*, *40*(7), 1364–1397. https://doi.org/10.1108/IJBM-08-2021-0371

Nyagadza, B., Muposhi, A., Mazuruse, G., Makoni, T., Chuchu, T., Maziriri, E. T., & Chare, A. (2024). Prognosticating anthropomorphic chatbots' usage intention as an e-banking customer service gateway: Cogitations from Zimbabwe. *PSU Research Review*, *8*(2), 356–372. https://doi.org/10.1108/PRR-10-2021-0057

Ofori, K. S., Anyigba, H., Adeola, O., Junwu, C., Osakwe, C. N., & David-West, O. (2022). Understanding post-adoption behaviour in the context of ride-hailing apps: The role of customer perceived value. *Information Technology & People*, *35*(5), 1540–1562. https://doi.org/10.1108/ITP-06-2019-0285

Okonkwo, C. W., Amusa, L. B., Twinomurinzi, H., & Fosso Wamba, S. (2023). Mobile wallets in cash-based economies during COVID-19. *Industrial Management & Data Systems*, *123*(2), 653–671. https://doi.org/10.1108/IMDS-01-2022-0029

Oliver, R. L. (1980). A cognitive model of the antecedents and consequences of satisfaction decisions. *Journal of Marketing Research*, *17*(4), 460–469. https://doi.org/10.2307/3150499

Parasuraman, R., & Riley, V. (1997). Humans and automation: Use, misuse, disuse, abuse. *Human Factors*, *39*(2), 230–253. https://doi.org/10.1518/001872097778543886

Park, G., Chung, J., & Lee, S. (2022). Effect of AI chatbot emotional disclosure on user satisfaction and reuse intention for mental health counseling: A serial mediation model. *Current Psychology (New Brunswick, N.J.)*, *42*(32), 1–11. https://doi.org/10.1007/s12144-022-03932-z

Rahi, S. (2017). Research design and methods: A systematic review of research paradigms, sampling issues and instruments development. *International Journal of Economics & Management Sciences*, *06*(02), 1–6. https://doi.org/10.4172/2162-6359.1000403

Rahi, S., Alghizzawi, M., & Ngah, A. H. (2023). Factors influence user's intention to continue use of e-banking during COVID-19 pandemic: the nexus between self-determination and expectation confirmation model. *EuroMed Journal of Business*, *18*(3), 380–396. https://doi.org/10.1108/EMJB-12-2021-0194

Reeves, B., & Nass, C. I. (1996). *The media equation: How people treat computers, television, and new media like real people and places*. Cambridge University Press.

Reis, H., & Shaver, P. (1988). *Intimacy as an interpersonal process* (pp. 367–389). Wiley. https://doi.org/10.4324/9780203732496-5

Rese, A., Ganster, L., & Baier, D. (2020). Chatbots in retailers' customer communication: How to measure their acceptance? *Journal of Retailing and Consumer Services*, *56*, 102176. https://doi.org/10.1016/j.jretconser.2020.102176

Rijsdijk, S. A., Hultink, E. J., & Diamantopoulos, A. (2007). Product intelligence: Its conceptualization, measurement and impact on consumer satisfaction. *Journal of the Academy of Marketing Science*, *35*(3), 340–356. https://doi.org/10.1007/s11747-007-0040-6

Rose, S., Clark, M., Samouel, P., & Hair, N. (2012). Online customer experience in e-retailing: An empirical model of antecedents and outcomes. *Journal of Retailing*, *88*(2), 308–322. https://doi.org/10.1016/j.jretai.2012.03.001

Roy, S. K., Shekhar, V., Lassar, W. M., & Chen, T. (2018). Customer engagement behaviors: The role of service convenience, fairness and quality. *Journal of Retailing and Consumer Services*, *44*, 293–304. https://doi.org/10.1016/j.jretconser.2018.07.018

Ruan, Y., & Mezei, J. (2022). When do AI chatbots lead to higher customer satisfaction than human frontline employees in online shopping assistance? Considering product attribute type. *Journal of Retailing and Consumer Services*, *68*, 103059. https://doi.org/10.1016/j.jretconser.2022.103059

Sands, S., Ferraro, C., Campbell, C., & Tsao, H.-Y. (2021). Managing the human–chatbot divide: How service scripts influence service experience. *Journal of Service Management*, *32*(2), 246–264. https://doi.org/10.1108/JOSM-06-2019-0203

Silva, S. C., De Cicco, R., Vlačić, B., & Elmashhara, M. G. (2023). Using chatbots in e-retailing – How to mitigate perceived risk and enhance the flow experience. *International Journal of Retail & Distribution Management*, *51*(3), 285–305. https://doi.org/10.1108/IJRDM-05-2022-0163

Song, S. W., & Shin, M. (2024). Uncanny valley effects on chatbot trust, purchase intention, and adoption intention in the context of E-commerce: The moderating role of avatar familiarity. *International Journal of Human–Computer Interaction*, *40*(2), 441–456. https://doi.org/10.1080/10447318.2022.2121038

Statista. (2023). Social Media in Vietnam – Statistics & Facts. Retrieved 18 October from https://www.statista.com/topics/8182/social-media-in-vietnam/#topicOverview

Steenkamp, J.-B. E. M., & Baumgartner, H. (2000). On the use of structural equation models for marketing modeling. *International Journal of Research in Marketing*, *17*(2-3), 195–202. https://doi.org/10.1016/S0167-8116(00)00016-1

Sung, J.-Y., Guo, L., Grinter, R. E., & Christensen, H. I. (2007). *"My Roomba Is Rambo": Intimate home appliances* (Ed.). UbiComp 2007: Ubiquitous Computing, Berlin, Heidelberg.

Trivedi, J. (2019). Examining the customer experience of using banking chatbots and its impact on brand love: the moderating role of perceived risk. *Journal of Internet Commerce*, *18*(1), 91–111. https://doi.org/10.1080/15332861.2019.1567188

van Doorn, J., Mende, M., Noble, S. M., Hulland, J., Ostrom, A. L., Grewal, D., & Petersen, J. A. (2017). Domo Arigato Mr. Roboto: Emergence of automated social presence in organizational frontlines and customers' service experiences. *Journal of Service Research*, *20*(1), 43–58. https://doi.org/10.1177/1094670516679272

Venkatesh, V. (2000). Determinants perceived ease of use: Integrating control, intrinsic motivation, and emotion into the technology acceptance model. *Information Systems Research*, *11*(4), 342–365. https://doi.org/10.1287/isre.11.4.342.11872

Xu, Y., Zhang, J., & Deng, G. (2022). Enhancing customer satisfaction with chatbots: The influence of communication styles and consumer attachment anxiety. *Frontiers in Psychology*, *13*, 902782. https://doi.org/10.3389/fpsyg.2022.902782

Yang, H., & Han, S.-Y. (2021). Understanding virtual reality continuance: An extended perspective of perceived value. *Online Information Review*, *45*(2), 422–439. https://doi.org/10.1108/OIR-02-2020-0058

Yang, H., & Lee, H. (2019). Understanding user behavior of virtual personal assistant devices. *Information Systems and e-Business Management*, *17*(1), 65–87. https://doi.org/10.1007/s10257-018-0375-1

Yoon, J., & Yu, H. (2022). Impact of customer experience on attitude and utilization intention of a restaurant-menu curation chatbot service. *Journal of Hospitality and Tourism Technology*, *13*(3), 527–541. https://doi.org/10.1108/JHTT-03-2021-0089

Zhang, J., & Curley, S. P. (2018). Exploring explanation effects on consumers' trust in online recommender agents. *International Journal of Human–Computer Interaction*, *34*(5), 421–432. https://doi.org/10.1080/10447318.2017.1357904